Regional Income Growth and House Price Dynamics in the Netherlands

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Set-up your environment

Regional Income Growth and House Price Dynamics in the Netherlands

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Tutorial group 3.3

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The Dutch housing market has been under pressure for several years now. Between 2013 and 2023 the average price of houses has more than doubled, while the income growth in the same period considerably lagged behind. The increasing gap between housing costs and purchasing power is not only an economic phenomenon, but also a social problem; it determines who does have access to a stable living situation and who doesn't.

It's striking that this development differs significantly across regions in the Netherlands. In the Randstad, prices increased much more than in the peripheral regions like Groningen or Zeeland, while the income development per region differs significantly too. Factors such as access to big employers, education level and local economic structure play an important role in this shift. And it is because of this regional variety that it's interesting to investigate the relation between income and house prices.

This project investigates the relationship between regional income growth and house price dynamics in Dutch provinces, based on CBS-data and the programming tool R. The goal is to discover in which regions the housing market detached most from the local income development.

Part 2 - Data Sourcing

2.1 Load in the data

we used 9 different data sets and combined we did this using this code for each first we cleaned the already existing datasets. aand combined everything together using this code: datacombined15tot23 <-

rbind(Data15161718, data2019, data2020, data2021, data2022, data2023).

joining

2.2 Provide a short summary of the dataset(s)

For this study two datasets from the Centraal Bureau voor de Statistiek (CBS) were used. The first dataset, Bestaande koopwoningen; verkoopprijzen prijsindex 2020=100 (85773NED), contains data on the variables ‘gemiddelde verkoopprijs van bestaande koopwoningen en bijbehorende prijsindex op provinciaal niveau’. The second dataset, inkomen van huishoudens; huishoudenskenmerken, regio (86004NED), contains information on the ‘gemiddelde inkomen van huishoudens, op provinciaal niveau’.

Both datasets are available on CBS StatLine. The dataset on the prices of houses spans from 2015 until 2025, the dataset on income only covers 2015 until 2023 (In order to see other years or regions, the filter option should be used to select the desired years). For the analysis in this research the overlapping years of 2015-2023 are studied.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

Part 3 - Quantifying

3.1 Data cleaning

Data cleaning datacombined15tot23goed. we mainly used this code to clean this data set: names(datacombined15tot23goed)[na
== “X1.000.euro...6”] <- “gem_st_inkomen_x_100” ETC.

Data cleaning woningprijsprov: we mainly used this code to clean this data set: woningprijsprov‘1995‘ $< -format(woningprijsprov$`1995`, big.mark = “.”, scientific = FALSE)

Please use a separate ‘R block’ of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1 creating new variable 1, we have created the average disposable income of each province over 8 years. using the data set: gem_inkomen_provincie, We used the following code for this: gem_inkomen_provincie <- aggregate(gem_bstb_inkomen_x_100 ~ Provincies, data = datacombined15tot23goed, FUN = mean)

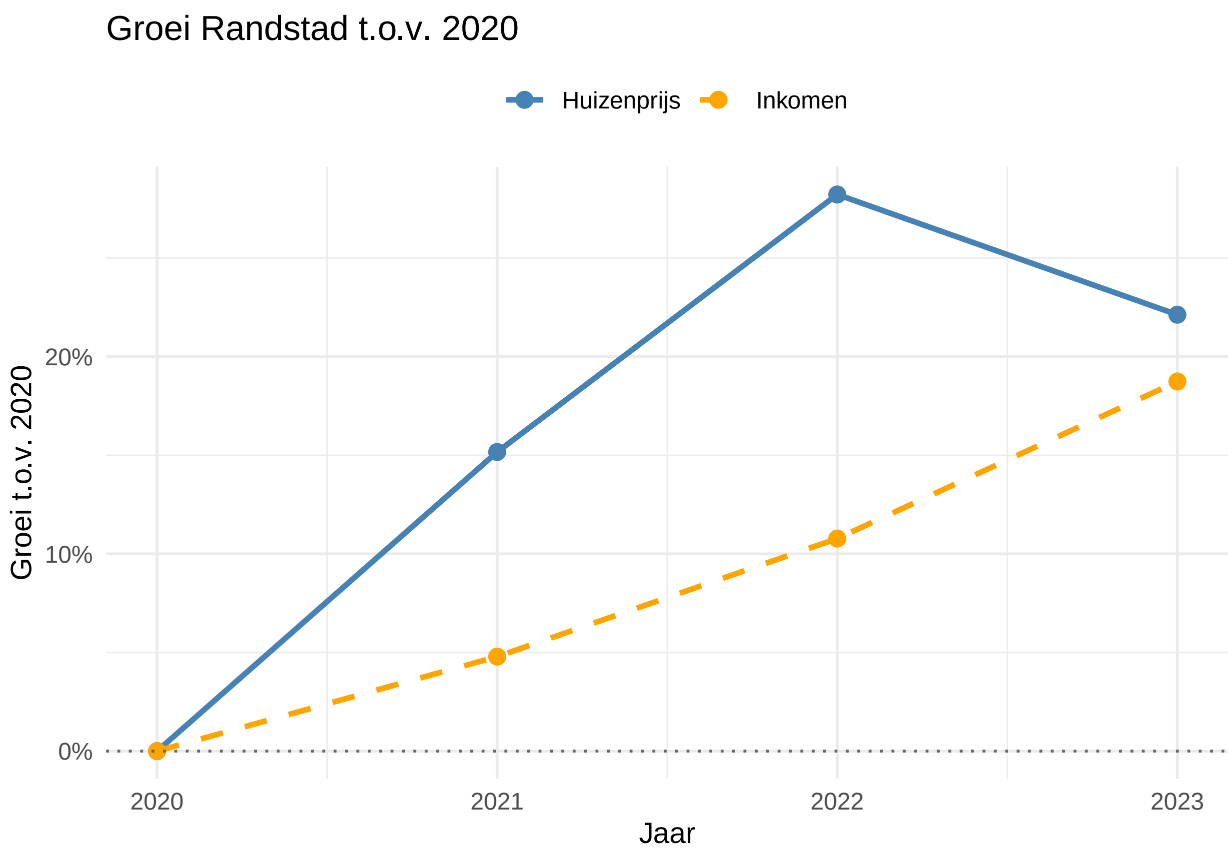
Variable 2 creating new variable 2 using the original dataset: datacombined15tot23goed, in which we represented the weighted averages of income for different provinces: inside and outside the Randstad.

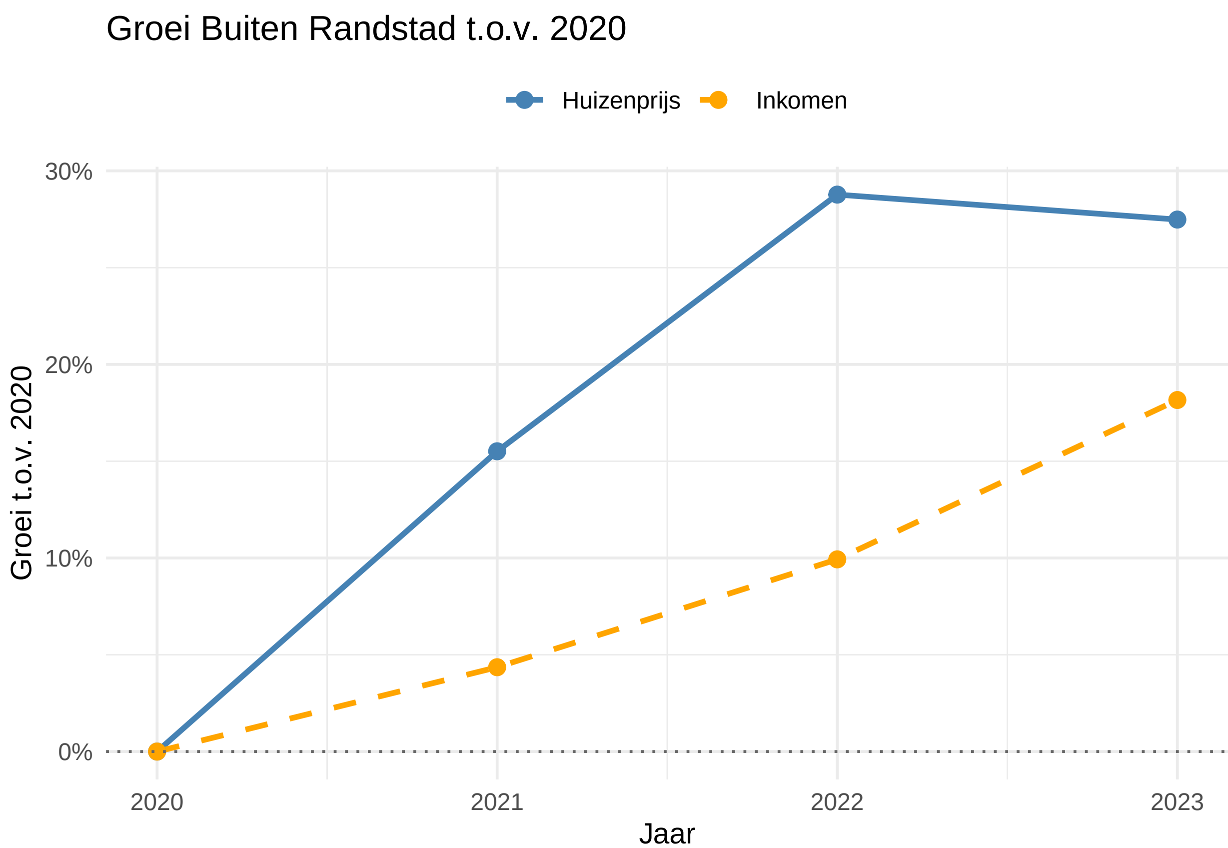
adjust the format of the current cleaned dataset in order to continue working with a renewed dataset, and also a renewed dataset, because for this new variable we actually need to merge our two datasets.

```r
woningprijsprov <- read.csv("goede data/goedehuizenprijsenperprov.csv", check.names = FALSE)

woningprijsprov <- woningprijsprov[!grepl("Bron", woningprijsprov$`Regio.s`), ]

woningprijsprov$Onderwerp <- NULL
woningprijsprov$`...4` <- NULL

woningprijsprov_lang <- reshape(woningprijsprov,
                                varying = c("2015", "2020", "2021", "2022", "2023"),
                                v.names = "gem_huizenprijs",
                                timevar = "jaar",
                                times = c(2015, 2020, 2021, 2022, 2023),
                                direction = "long")

woningprijsprov_lang$id <- NULL

write.csv(woningprijsprov_lang, "goede data/goedehuizenprijsenperprov_lang.csv", row.names = FALSE)
```

3.3 Visualize temporal variation

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##     discard

## The following object is masked from 'package:readr':
##
##     col_factor
```


Groei Randstad t.o.v. 2020
Huizenprijs
Inkomen
Groei t.o.v. 2020
20%
10%
0%
2020
2021
2022
2023
Jaar



3.4 Visualize spatial variation

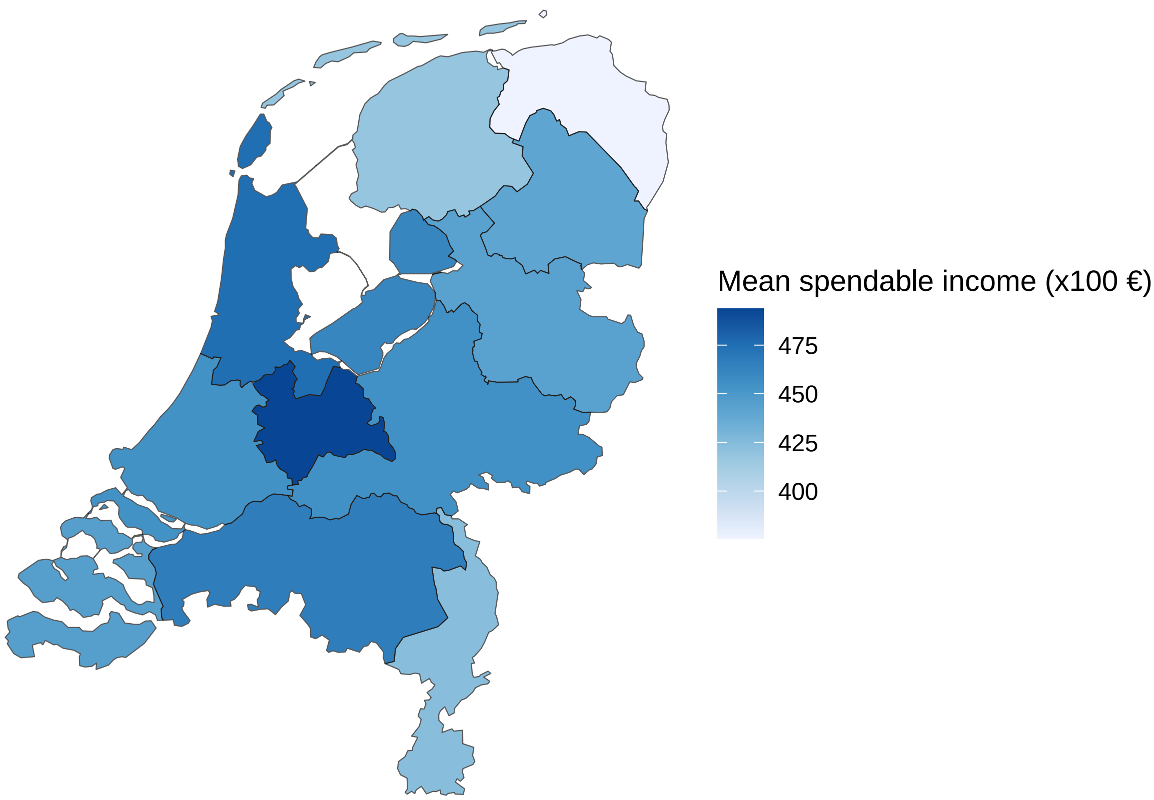
```
## The following package(s) will be installed:
## - ggplot2 [4.0.3]
## - sf      [1.1-1]
## These packages will be installed into "C:/Github opdracht/R-project-group-3-3/renv/library/windows/R-
##
## # Installing packages --------------------------------------------------------
## - Installing sf ...                             OK [linked from cache]
## - Installing ggplot2 ...                        OK [linked from cache]
## Successfully installed 2 packages in 24 milliseconds.
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
## Reading layer ‘provincie_2024’ from data source
##   ‘https://cartomap.github.io/nl/wgs84/provincie_2024.geojson’
##   using driver ‘GeoJSON’
## Simple feature collection with 12 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 3.358 ymin: 50.751 xmax: 7.218 ymax: 53.554
## Geodetic CRS:  WGS 84
```

```
##  [1] "Groningen"     "Fryslân"       "Drenthe"       "Overijssel"
```

```
##  [5] "Flevoland"     "Gelderland"    "Utrecht"       "Noord-Holland"
##  [9] "Zuid-Holland"  "Zeeland"       "Noord-Brabant" "Limburg"
```




```
## <ggplot2::labels> List of 2
##  $ title   : chr "Mean Spendable Income per Province"
##  $ subtitle: chr "Average over 2015-2023"

## Reading layer ‘provincie_2024’ from data source
##   ‘https://cartomap.github.io/nl/wgs84/provincie_2024.geojson’
##   using driver ‘GeoJSON’
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```

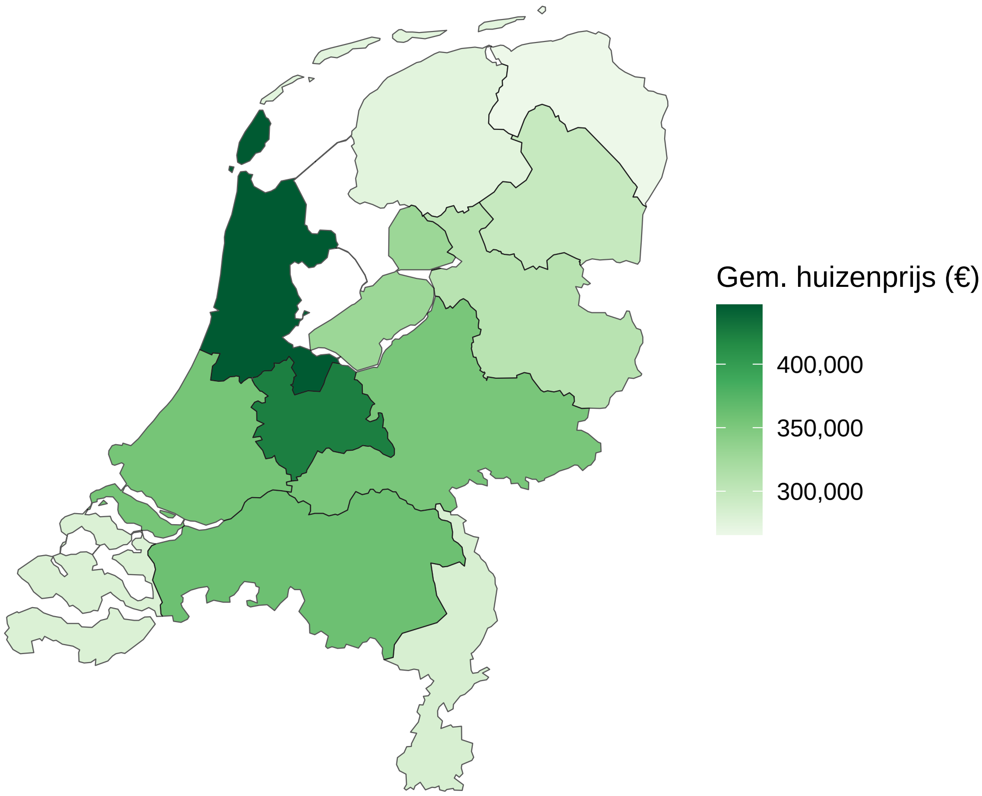


```
## <ggplot2::labels> List of 2
##  $ title   : chr "Gemiddelde Huizenprijs per Provincie"
##  $ subtitle: chr "Gemiddelde over de laatste jaren"
```

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.